Rational curiosity predicts the dynamics of habituation and dishabituation in infants and adults

Your Name

2023-11-13

Contents


```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.4
## v forcats   1.0.0     v stringr   1.5.1
```

```
## v ggplot2   3.5.0     v tibble    3.2.1
## v lubridate 1.9.2     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(here)
```

```
## here() starts at /Users/galraz/Developer/pkbb_paper_writing
```

```r
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(lmerTest)
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```r
library(broom.mixed)
library(magrittr)
```

```
##
## Attaching package: 'magrittr'
##
## The following object is masked from 'package:purrr':
##
##     set_names
##
## The following object is masked from 'package:tidyr':
##
##     extract
```

```r
library(pbkrtest)
```

```r
#Load packages & datasets
library(tidyverse)
library(here)
library(lme4)
library(lmerTest)
library(broom.mixed)
library(magrittr)
library(pbkrtest)

infant_d <- read_csv(here("data/infants/infant_processed_data.csv"))
```

```
## New names:
## Rows: 1162 Columns: 13
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (5): experiment, fam_or_test, unique_id, test_type, sex dbl (7): ...1,
## subject_num, animal, block_num, fam_duration, LT, age lgl (1): exclude
## i Use ‘spec()‘ to retrieve the full column specification for this data. i
## Specify the column types or set ‘show_col_types = FALSE‘ to quiet this message.
## * ‘‘ -> ‘...1‘
```

```r
# Load models
model_1a <- readRDS(here("cached_models/infants/model_1a.rds"))
model_2a<- readRDS(here("cached_models/infants/model_2a.rds"))
model_2b<- readRDS( here("cached_models/infants/model_2b.rds"))
model_2c <- readRDS( here("cached_models/infants/model_2c.rds"))
model_2d <- readRDS(here("cached_models/infants/model_2d.rds"))
model_3a <- readRDS(here("cached_models/infants/model_3a.rds"))
model_3b <- readRDS(here("cached_models/infants/model_3b.rds"))
model_3c <- readRDS( here("cached_models/infants/model_3c.rds"))
```

1 Abstract

How do we decide what to look at and when to stop looking? Even very young infants engage in active visual selection, looking less and less as stimuli are repeated (habituation) and regaining interest when novel stimuli are subsequently introduced (dishabituation). The mechanisms underlying these looking time changes remain uncertain, however, due to limits on both the scope of existing formal models and the empirical precision of measurements of infant behavior. We develop the Rational Action, Noisy Choice for Habituation (RANCH) model, which operates over raw images and makes quantitative predictions of participants' looking behaviors. In a series of pre-registered experiments, we measure infants' and adults' looking time across a range of stimulus exposure durations for both repeated and novel stimuli and show that they are predicted by RANCH. Our model also makes out-of-sample predictions about the magnitude of adult and infant dishabituation to a range of novel stimuli in an independent experiment. By framing looking behaviors as rational decision-making, this work identifies how the dynamics of learning and exploration guide our visual attention from infancy through adulthood.

2 Introduction

From birth, humans engage in active learning. Even before they gain independent mobility, infants make choices about what to look at and when to stop looking [13, 27]. Developmental psychologists have both studied this process of visual decision-making and also relied on it as a key methodological tool, particularly two key phenomena: habituation and dishabituation. Habituation occurs when infants show reduced interest upon repeated exposure to the same stimulus, whereas dishabituation entails renewed interest following the subsequent introduction of a novel stimulus. Dishabituation, also known as novelty preference, is often leveraged to infer infants' perceptual and cognitive abilities [2, 3, 11]: in order for an infant to dishabituate, they must distinguish between the original stimulus and the novel one. Despite the robust documentation of habituation and dishabituation in the literature, the mechanisms underlying these changes in gaze duration remain poorly understood. In this paper, we bridge this gap by introducing a rational model of habituation and dishabituation. We show that this rational model makes quantitative predictions of looking times toward different stimuli.

An important early conceptual model of infant looking time posited that habituation and dishabituation are influenced by the amount of information to be encoded by the infant from the stimulus [17]. Initially, infants look longer at stimuli containing substantial unprocessed information, showing a familiarity preference (Fig. 1A). As exposure accumulates, less information remains unprocessed, leading to shorter looking time at the original stimulus and by comparison, relatively longer looking time to a new stimulus (i.e. novelty preference). While this model has had significant influence, the absence of specifications regarding the encoding process or a quantitative definition of information to be encoded have permitted post-hoc interpretations of looking time measurements. For instance, when infants are looking longer at stimulus A than stimulus B, researchers could interpret the looking time difference as either a novelty preference or a familiarity preference. This interpretive ambiguity has spurred repeated concerns regarding whether looking time measurements should form the basis for key assertions in developmental psychology [4, 14, 24]. In other words, developmental psychology needs formal models that make quantitative predictions, but the prevailing model proposed by Hunter and Ames [17] can not provide the necessary clarity to support robust inferences.

In contrast, more recent computational models aim to formalize the mechanisms underlying looking time changes. One family of models characterize infants' looking behaviors through information-theoretic metrics derived from ideal observer models: the models acquire probability distribution from event sequence and derive information-theoretic measures from the ideal learner's belief before and after each event [19, 25, 26]. For instance, Kidd, Piantadosi, and Aslin [19] developed a model that used surprise to quantify the extent to which the model learned from each observation. This measure correlated well with infants' looking behavior via a U-shaped curve: infants were least likely to look away when the surprisal of the event was neither too high nor too low (1B).

While these models provide quantitative fits to variation in looking time, they are not models of how an agent makes the decisions of whether to keep looking at the same stimulus or to look at something else. Instead, these models describe correlations between model-derived, information-theoretic measures and infant overall looking towards whole sequences of events. Relatedly, these models do not explain why infants would ever look at one static stimulus for a long time, rather than encode it instantaneously – there is no accounting of perceptual noise that needs to overcome by taking multiple perceptual samples that would justify why looking extends over time [c.f. 5, 18]. Owing to this constraint, these models can only be used to predict infant behavior in a specific paradigm: looking to a continuous stream of discrete stimuli. This paradigm is a significant deviation from the more common looking time paradigm in which the participants gradually encoded individual stimulus items in a self-paced manner. It is unclear whether the success of these quantitative models can be generalized to the more common looking time paradigms.

To address these issues, we developed the Rational Action, Noisy Choice for Habituation (RANCH) model [6]. RANCH describes an agent's looking behavior as rational exploration based on a sequence of noisy perceptual samples. The model construes the looking time paradigm as a series of binary decisions: to keep sampling from the current stimulus, or to look at anything else. The model makes sampling decisions based on the Expected Information Gain (EIG) of the upcoming perceptual sample, choosing to keep looking or look away based on which one would in expectation yield the most information; it therefore can be seen as a rational analysis of looking behavior [21, 22]. Rational analyses of behavior model decision-making as optimally adapted to the environment, given specific constraints. Identifying a behavior as rational provides a framework for predicting and influencing future actions, and facilitates the development of formal connections between human behavior and the environment in which it occurs [1].

Previously, we validated this rational analysis using a large dataset of adults engaging in a self-paced looking paradigm. We asked adults to look at sequences of stimuli consisting of a repeating familiar background stimulus (resulting in habituation), and a single, novel deviant stimulus after varying familiarization durations (resulting in dishabituation). Effects of prior exposure and novelty on adults' looking time were well captured by RANCH [6].

Here, we ask how well RANCH describes the dynamics of infant habituation and dishabituation, and compare it to other classic models of infant attention. Figure 1C shows schematic predictions of RANCH for the experimental setting described above: variable prior exposure to a familiar stimulus, followed by a novel or a familiar simulus. Rational exploration of a simple perceptual concept results in a monotonic decrease of attention to familiar items as a function of prior exposure.

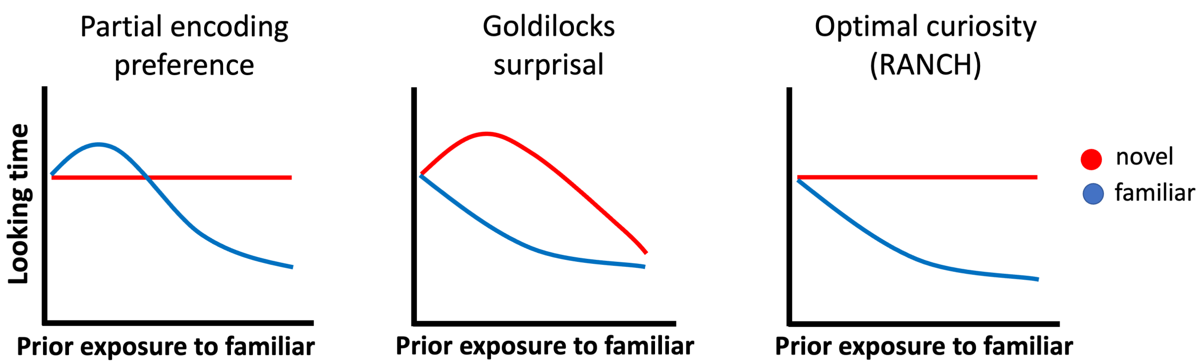

Figure 1: Conceptual models of looking to novel vs. familiar items according to three models of infant attention: A) The partial encoding model proposed by Hunter & Ames (1988) which suggests a shift from familiarity to novelty preferences, B) The Goldilocks model proposed by Kidd et al. (2012) which suggests that infants prefer to attend to intermediately surprising events, and C) the 'optimal curiosity' model introduced by Cao et al. (2023) which suggests that infants are maximizing expected information gain from noisy perceptual samples.

One critical challenge for differentiating formal theories of looking time in infants is that looking time measurements are noisy, and many infant experiments provide only limited precision in their estimates of key quantities [7, 8]. In the current work, we develop a novel, within-subject experimental design that allows us to sensitively measure infants' looking time in experiments. Beyond validating RANCH's predictions in infants, this paradigm enables us to conduct interpretable developmental comparisons of parameter fits between infants and adults – since the experimental design was analogous to each other – thereby characterizing how the mechanisms underlying looking time may change over development.

In addition, we incorporated recent progress in convolutional neural networks (CNNs) to allow RANCH to learn from raw images. Previous models have relied on using arbitrary representations of stimuli [e.g. object location, 19, hand-specified binary feature vector, 6]. However, recent CNNs have offered insights into how the visual system encodes objects [10, 16, 30]. The activations of these brain-inspired neural networks form embedding spaces, each of which can be seen as a quantitative hypothesis about how humans represent visual stimuli [28]. In RANCH, we leveraged these principled stimulus representations to enable RANCH to learn from raw images. In other words, RANCH can 'see' an experiment in a way similar to the one infants saw, and make predictions for previously unseen experiments using novel stimuli.

Last but not least, while previously we have shown that RANCH can successfully model habituation and dishabituation, we now test RANCH's ability to predict responses in new data. To conduct these tests, we first fit RANCH's parameters to a training data set from the habituation-dishabituation experiment in which participants saw sequences of stimuli. Then, we use the best-fitting parameters to generate predictions for a new experiment designed to measure subtle differences in dishabituation magnitude based on stimulus similarity. In this experiment, we systematically varied the similarity between habituation and dishabituation stimuli such that dishabituation stimuli differed in their pose angle, their number, their identity, or their animacy. This experiment tests a prediction from Hunter and Ames [17] model of habituation and dishabituation: that observers' dishabituation magnitude should be related to the similarity between the habituated stimulus and the novel stimulus. The more dissimilar two stimuli are, the more one should dishabituate to the novel stimulus.

The goal of this paper is to investigate the computational mechanisms underlying habituation and dishabituation by testing to what extent these behaviors can be explained by a rational analysis of looking time. To preview our results, we found support for the view that habituation and dishabituation are driven by rational curiosity by showing that RANCH predicts looking time in adults and infants. Furthermore, we also tested the importance of individual assumptions RANCH made via target ablation simulations. Removal of components of perceptual representation, learning progress, and decision making all result in substantially worse fit to data. Finally, we generate predictions for the novel task in which we vary the similarity between the habituation and dishabituation stimulus, and find that RANCH also captures the particular ordering of the dishabituation magnitude as a function of stimulus dissimilarity, thereby predicting a novel qualitative phenomenon (graded dishabituation) without ever being trained on it.

3 Rational Action, Nosy Choice for Habituation model

Rational Action, Noisy Choice for Habituation model (RANCH) is a Bayesian perception and action model in which a learner makes optimal perceptual sampling decisions [6]. In the current implementation, we conceptualized the learning problem in habituation as learning a visual concept – a probabilistic representation of the stimuli it experiences. The model achieves learning through observing a series of noisy perceptual samples from a sequence of stimuli. During the learning process, the model makes decisions about how many samples to receive of each stimulus before moving to the next stimulus. The number of samples that the model takes on each stimulus is then used as a proxy for looking time duration at each sample.

RANCH provides a probabilistic framework for making perceptual decisions. In this section, we will describe the three modular components RANCH includes: the perceptual representation, learning model, and decision model.

3.1 Perceptual representations

We use the deep convolutional neural network (CNN) ResNet-50 to encode the stimuli used in behavioral experiments. ResNet-50 is an architecture that has been widely used in computer vision [15]. A total of 50 layers are implemented in this architecture: it starts with a convolution layer, includes several groups of residual blocks that help preserve information through the network, and ends with a global average pooling layer and a fully connected layer for classification. We used a version of ResNet-50 that was pretrained on ImageNet [9]. Training on ImageNet optimizes the Resnet-50's activations for image classification, and we can use these activations as an perceptual "embedding space". When we pass new stimuli through a pretrained network, we extract these activations to obtain principled embeddings of stimuli. We used the first three principal components of the embedding space to limit the dimensionality of the downstream learning problem.

ResNet-50 is by no means the only possible CNN for deriving perceptual representations. We also explored alternative models, such as a CNN trained on SayCam data [23] or a CNN whose representations were aligned to human behavior [20]. However, we did not find substantial differences between the embedding spaces provided by ResNet-50 and other (often more complicated) models (See SI).

3.2 Learning model

In the current instantiation, RANCH models looking behaviors in habituation experiments as Bayesian concept learning [12, 29]. The model learns the concept through a series of noisy perceptual samples taken from a stimulus. The concept to be learned is parameterized by μ, σ, which represents beliefs about the location and variance of the presented concept in the embedding space. The concept μ, σ generates exemplars y, which is equivalent to the stimulus seen by the human participants. But RANCH does not directly learn from the stimulus. Instead, it learns from the series of noisy perceptual samples $\bar{z}$ generated by the exemplar. The noisy perceptual sample is corrupted by zero-mean noise, represented by ϵ. A plate diagram is shown in Figure 2B. At each time step, the model takes one perceptual sample to update the prior following Bayes Rule:

$$p(\mu, \sigma|\bar{z}) = \frac{p(\bar{z}|\mu, \sigma) * p(\mu, \sigma)}{p(\bar{z})}$$

3.3 Decision model and linking hypothesis

The Bayesian learning model did not address how the model decides to keep sampling the same stimulus or to move on to the next stimulus. We enable RANCH's decision-making process by calculating the expected information gain (EIG) of the next perceptual sample, which we define as the average KL-divergence across a grid of possible next observations. EIG is a metric used under the rational analysis of information-seeking behavior [1, 22]. RANCH then uses a softmax choice (with temperature = 1) between the information theoretic measures of the next sample and a constant "environmental EIG". The constant here is assumed to be the amount of information gain provided by the environment when the learner looks away from the stimulus.

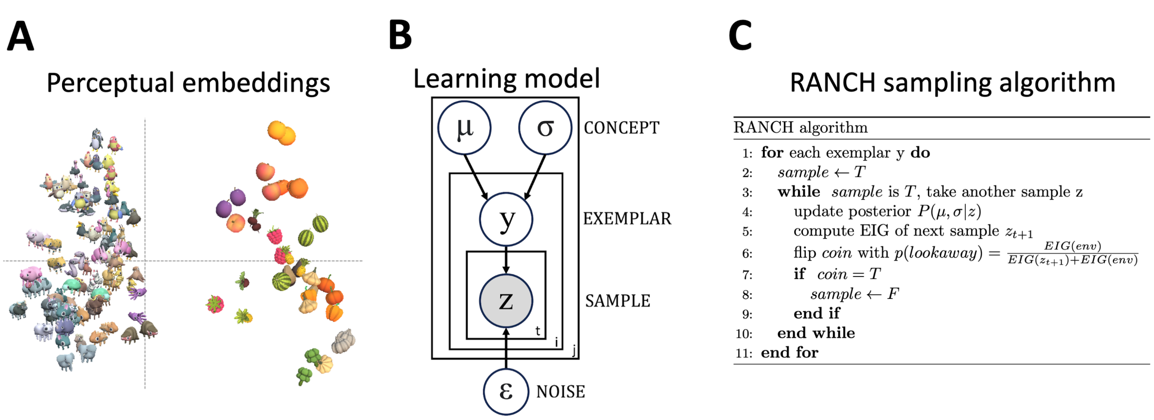

Figure 2: Three components of RANCH: A) Perceptual representation: Obtaining stimulus embeddings from ResNet-50 final layer activations, B) Learning model: Plate diagram for Gaussian concept learning from noisy observations, C) Decision model: RANCH repeatedly samples until environmental EIG outweighs EIG from another sample.

3.4 Implementation

RANCH has a series of parameters: the learner's priors over the concept to be learned (μ, σ), the learner's priors over their perceptual noise (ϵ) and the environmental EIG (EIG_{env}). We conducted an iterative grid search over these parameters.

Given that the element of accounting for the learner's own noise breaks the conjugacy of updating the concept mean and variance, we resorted to grid approximation to perform the sample-by-sample Bayesian updates. We discretized the grids over μ, $sigma$, y (exemplars) and z (noisy samples) to perform inference, as well as the grid of possible next observations when computing EIG.

To average out noise resulting from grid approximation, we ran each combination of parameter setting and trial type 100 times. To avoid bias, these 100 simulations used 20 different "jitters" - such that the exact location of the points in the discrete grids was randomly chosen. We used the number of samples the model decided to take from each stimulus as a proxy for looking time, effectively converting the binary choice of "look" or "look away" to a continuous variable. We then average the number of samples across the 1000 runs.

This computationally intensive process required significant resources. The grid approximation, combined with repeated simulations, demanded high-performance cluster GPUs to perform vectorized computation using a PyTorch implementation of our model. Running a single parameter setting through an experiment took approximately 10 hours on a GPU - by distributing different parameter settings across GPUs, a full parameter search took about 3 days of compute.

4 RANCH predicts habituation and dishabituation in relation to exposure duration

A key difference between the models in Figure 1 is their prediction with regard to the effect of familiarity on looking time. To address this, we collected looking time data from both infants and adults in an experiment where we varied exposure duration to a familiar stimulus, and measured subsequent looking time to a novel vs. familiar stimulus. We then let RANCH "see" the same stimuli and procedure used in our behavioral experiments. We then evaluated the model's performance against the behavioral datasets.

4.1 Behavioral experiments

4.1.1 Infant experiment

We tested infants in a novel experimental paradigm in which exposure duration to the stimuli were manipulated in a within-subject design 3 (Experiment 1). Each infant saw six blocks. In each block there was an familiarizaiton phase and a test trial. During the familiarization, infants were presented with one stimulus, repeated 0 to 9 times. After the familiarization, infants' looking time was measured on a test trial. The test trial showed either the same stimulus as the preceding trials, or a new stimulus. We ran three versions of this experiment, each with three different familiarization durations. We tested a combined sample of 103 7-10 month old infants, with 310 infants in Exp 1A (0, 4 or 8 repetitions), 408 infants in Exp 1B (1, 3 or 9 repetitions) and 444 infants in Exp 1C (2, 4 or 6 repetitions; total n=103 (M_{age} = 9.38 months, 47 female).

4.1.2 Adult experiment

We ran a self-paced looking time paradigm experiment on Prolific, collecting data from XXXX adult participants (FIGURE). The stimuli used in the adult experiment are drawn from the same stimuli set used in the infant experiment. During the experiment, participants were asked to watch sequences of animations at their own pace, and they could press a key to proceed to the next trial whenever they wanted to. The time between the onset of the stimulus and the key press was used as a proxy for looking time. Each sequence of animations has two stimuli: the familiar stimulus and the novel stimulus. The familiar stimulus repeated over and over again within a sequence, whereas the novel stimulus only appeared once and it appeared at either the second, the fourth, or the sixth position in the sequence. Modeling Procedure

The overall modeling strategy is for RANCH to ‘see’ the experiment the same way as the participants do. RANCH learned from the embeddings derived from the stimuli used in the experiments, and the modeling procedure mirrored the experimental set-up used in the infants experiment and adult experiment respectively (Fig. 2).

To simulate the infant experiment, the model was first presented with a series of repeating stimuli. For each stimulus, the model took 5 samples each and then moved on to the next stimulus. This phase is similar to the exposure phase infants went through, since during the exposure phase infants had no control over the progression of the stimuli sequence. After the familiarization sequence, the model was presented with the test trial, which either has a novel stimulus or a familiar stimulus (i.e. the repeated stimulus) . The number of samples the model took on this test trial was the “looking time” that the model produced.

In contrast, to simulate the adult experiment, RANCH makes decisions on when and whether to progress to the next trial in the sequence. At each animation in the sequence, the model takes a certain number of samples before deciding to move on to the next animation, until the end of the sequence. Like the adult experiment, the sequence has a familiar stimulus that repeats throughout the sequence, and a novel stimulus that appears once. The model predicts looking times to each stimulus in the sequence.

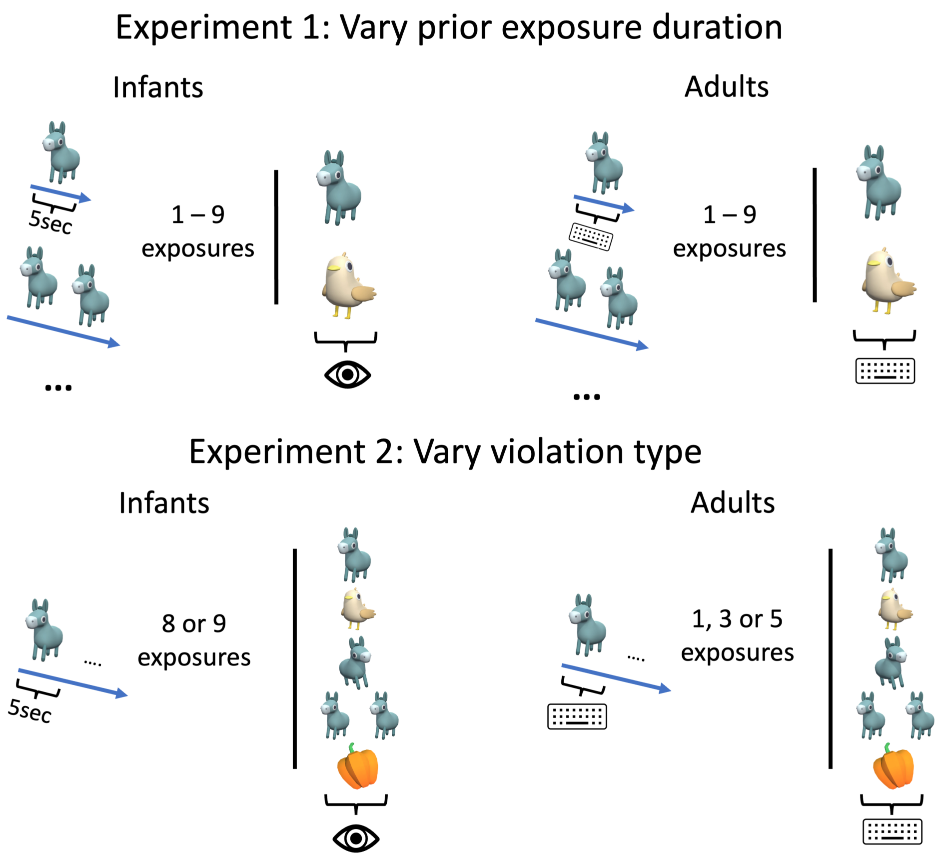

Figure 3: Experimental deisgn

##Evaluating model with behavioral results

The paradigm successfully captured habituation in both infants and adults. In infants, as the number of trials in exposure increased, the looking time at the familiar stimulus in the test trials decreased (STATS). We also found evidence for infants’ dishabituation: looking time for the novel test trials was longer than the familiar trial (STATS). Likewise, we also found that adult experimental paradigm successfully captures habituation and dishabituation (STATS).

RANCH predicted habituation and dishabituation in both infants and adults. We evaluated each parameter set's performance by running a K-fold validation test. At each fold, we fit a linear model predicting the looking time data in the fold. We then used the coefficients and the intercepts from the fitted model to scale the model results to the rest of the behavioral data. We compute RMSE and the Pearson's r between the scaled model results and the test set. This process is repeated for each fold and for each parameter setting. For infants, we used K = 2 due to the data sparsity. For adults, we used K = 10. The K-fold validation suggests that our model performance is largely robust across parameters (Table ??).

##Comparison with alternative linking hypotheses We also compared the performance of RANCH when using EIG, which is the fully rational linking hypothesis, to other, proxy-based linking hypothesis used in prior literature: Kullback-Leibler divergence (KL-divergence) and surprisal. KL-divergence measures the information gained when one's beliefs are updated, effectively quantifying 'learning progress', while surprisal captures how inconsistent incoming information is with prior expectations. Both have been shown to be associated with infants' looking behaviors (CITE, CITE). We then compared the model's performance based on the parameter that provides the best fits to the full dataset. Out of the three linking hypotheses, we found that EIG and KL-divergence were comparable to each other (Infants: STATS; Adults: STATS). But surprisal-based RANCH performed worse in both populations (Infants: STATS; Adults: STATS). We also found that the noisy perception assumption and the learning assumptions were both critical to RANCH's performance.

##Comparison with lesioned model

We also examined the importance of RANCH's key assumption. To do so, we created two lesioned models: one model assumes each perceptual sample is noiseless ("No noise model"), and the other model assumes the sampling decision was random instead of based on the learning model ("No learning model"). Both lesioned models provided poor fits for the behavioral datasets (STATS).

##Parameter Interpretation / Developmental Comparison One advantage of RANCH is the interpretability of its parameters. Beyond finding that RANCH's fit to the data is robust across parameters, we can also ask which specific parameters maximize that fit. In particular, comparing the best fitting parameters between infants and adults could provide insights into the origin of the developmental differences in behavior (see 'Joint Scaling' under Methods for technical details). When investigating which parameters were most influential in achieving better fit, we found that sd_epsilon played a key role in improving fit. Conceptually, sd_epsilon represents an agent's beliefs about how much noise there is in their own perceptual input. We found that this value tends to be high for infants, and low for adults ([SUPPLEMENTARY FIGURES WITH HISTOGRAM OF RMSE COLORED BY PARAMETER VALUES]).

##Discussion Through within-subjects measurements of looking time, we examined the sensitivity of looking to familiar vs. novel stimuli as a function of prior exposure. We did not find evidence for non-linearities predicted by prior models (Fig. 1A & 1B), and instead found that habituation and dishabituation followed monotonic trends as predicted by RANCH (Fig. 1C).

We next turn to test RANCH's ability to make predictions in a novel dataset.

4.2 Methods

5 {r child = "04_experiment2_methods.Rmd"}

5.1 Results

6 {r child = "05_experiment2_results.Rmd"}

7 Discussion

8 {r child = "06_general_discussion.Rmd"}

9 References

References

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